

34-2c A Formula for the Spending Multiplier

Some simple algebra allows us to derive a formula for the size of the multiplier effect that arises when an increase in government purchases induces increases in consumer spending. An important number in this formula is the *marginal propensity to consume* (*MPC*)—the fraction of extra income that a household consumes rather than saves. For example, suppose that the marginal propensity to consume is $\frac{3}{4}$. This means that for every extra dollar that a household earns, the household spends \$0.75 ($\frac{3}{4}$ of the dollar) and saves \$0.25. With an *MPC* of $\frac{3}{4}$, when the workers and owners of Boeing earn \$20 billion from the government contract, they increase their consumer spending by $\frac{3}{4} \times \$20$ billion, or \$15 billion.

To gauge the impact of a change in government purchases on aggregate demand, we follow the effects step-by-step. The process begins when the government spends \$20 billion and, as a result, increases national income (earnings and profits) by the same amount. With an extra \$20 billion of income, consumers increase spending by $MPC \times \$20$ billion. This additional consumer spending raises the income for the workers and owners of the firms that produce the consumption goods by the same amount. With this second increase in income, consumers increase spending again, this time by $MPC \times (MPC \times \$20 \text{ billion})$. These feedback effects go on and on.

To determine the total impact on the demand for goods and services, we add up all these effects:

Change in government purchases =	\$20 billion
First change in consumption	= $MPC \times \$20 \text{ billion}$
Second change in consumption	= $MPC^2 \times \$20 \text{ billion}$
Third change in consumption	= $MPC^3 \times \$20 \text{ billion}$
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.	.
.	.
Total change in demand	
	= $(1 + MPC + MPC^2 + MPC^3 + \dots) \times \20 billion .

Here “...” represents an infinite number of similar terms. Thus, we can write the multiplier as follows:

$$\text{Multiplier} = 1 + MPC + MPC^2 + MPC^3 + \dots$$

This multiplier tells us the demand for goods and services that each dollar of government purchases generates.

To simplify this equation for the multiplier, recall from math class that this expression is an infinite geometric series. For x between -1 and $+1$,

$$1 + x + x^2 + x^3 + \dots = 1/(1 - x).$$

In our case, $x = MPC$. Thus,

$$\text{Multiplier} = 1/(1 - MPC).$$

For example, if MPC is $3/4$, the multiplier is $1/(1 - 3/4)$, which is 4. In this case, the \$20 billion of government spending generates \$80 billion of demand for goods and services.

This formula for the multiplier shows that the size of the multiplier depends on the marginal propensity to consume. While an MPC of $3/4$ leads to a multiplier of 4, an MPC of $1/2$ leads to a multiplier of only 2. Thus, a higher MPC means a larger multiplier. To see why, remember that the multiplier arises because higher income induces greater consumer spending. The higher the MPC , the more consumption responds to a change in income, and the larger the multiplier.